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Introduction to Inverse Problems (02624)

Week 2

Inverse and Ill-posed problems

Concerns the determination of a phenomenon from indirect measurements

Given forward model F and measurements y, recover the phenomena x from

$$F(x) = y.$$

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Three questions for well-posedness according to Hadamard (1923)

- Existence
- Uniqueness
- Stability

If any fails, the problem is *ill-posed*.

Compact operators - integral operators

Many inverse problems come in the form

$$Kx = y$$

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Example 1: Integral operator

$$Kx(t) = \int_a^b k(t,s)x(s)\mathrm{d}s$$

is compact in $L^2(a,b)$, e.g., if $k \in L^2((a,b)^2)$.

Example 2: A matrix $A \in \mathbb{R}^{n\times m}$ provides the compact operator

$$\mathbb{R}^m \ni x \mapsto Ax = y.$$

Model problem 0: Differentiation

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Inverse problem: given a function y find the derivative $y' = x$.

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Note that

$$Kx(t) = \int_0^t x(s)\mathrm{d}s = \int_0^1 k(t,s)x(s)\mathrm{d}s, \qquad k(s,t) = \begin{cases} 1 & s < t, \\ 0 & s > t. \end{cases}$$

Model problem 1: Abel integral

From Computerized Tomography in radially symmetric case:
Abel integral operator

$$Kx(t) = \int_0^1 k(t,s)x(s)\mathrm{d}s, \quad t > 0, \qquad k(t,s) = \begin{cases} \frac{1}{\sqrt{(t-s)}}, & s < t \\ 0, & s > t. \end{cases}.$$

Inverse problem: from $y = Kx$, find y.

Model problem 2: Backwards Heat Equation

$$\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = 0, \qquad 0 < x < \pi, \quad t > 0$$

Boundary conditions: $u(0, t) = u(\pi, t) = 0$

Initial condition: $u(x, 0) = f(x)$.

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Forward problem: from f to terminal temperature $g = u(\cdot, T)$

$$g(x) = Kf(x) = \int_0^\pi k(x,y) f(y)\,\mathrm{d}y, \qquad k(x,y) = \frac{2}{\pi} \sum_{n=1}^{\infty} e^{-Tn^2} \sin(nx) \sin(ny).$$

Inverse problem: Go back in time from g to f.

Model problem 2: Backwards Heat Equation

With x the initial heat distribution, y the terminal, s, t space variables,

$$y(t) = Kx(t) = \int_0^{\pi} k(t, s)x(s)\, \mathrm{d}s$$

with

$$k(t, s) = \frac{2}{\pi} \sum_{n=1}^{\infty} e^{-Tn^2} \sin(nt) \sin(ns).$$

Inverse problem: Go back in time from y to x.

Model problem 3: Deblurring

Define

$$k(z) = \begin{cases} 1 + \cos(\pi z/3), & |z| < 3, \\ 0, & |z| \geq 3. \end{cases}$$

The problem is then defined via the operator K in $L^2(-6, 6)$

$$Kx(t) = \int_{-6}^{6} k(s-t)x(s)\mathrm{d}s. \tag{1}$$

Discretization of integral equation

Operator equation vs matrix equation

$$Kx(t) = \int_a^b k(t,s)x(s)\,\mathrm{d}s = y(t) \qquad \rightarrow \qquad Ax = y$$

Quadrature + collocation:

$$\int_a^b k(t,s)x(s)\,\mathrm{d}s \approx \sum_{j=1}^{J} w_j k(t,s_j)x(s_j),$$

$$\sum_{j=1}^{m} w_j k(t_i,s_j)x(s_j) = y(t_i), \quad i = 1,2,\ldots n.$$

That is $Ax = y$ with

$$A_{ij} = w_j k(t_i,s_j),\ y_i = y(t_i),\ x_j:\ \text{approximation of samples } x(s_j).$$

Discretization of integral equation

$$Kx(t) = \int_a^b k(t,s)x(s)\mathrm{d}s = y(t) \qquad \rightarrow \qquad Ax = b$$

Expansion method (Galerkin):
Consider for two basis (for finite dimensional subspaces of $L^2(a,b)$) $\{\phi_j\}$ and $\{\psi_i\}$ and approximate

$$x \in \mathrm{span}(\phi_1, \phi_2, \dots, \phi_m), \qquad y \in \mathrm{span}(\psi_1, \psi_2, \dots, \psi_n).$$

Requiring the *residual* $Kx - y$ to be orthogonal to $\mathrm{span}(\psi_1, \psi_2, \dots, \psi_n)$

$$A_{ij} = \int\int \psi_i(t)k(t,s)\phi_j(s)\,\mathrm{d}s\mathrm{d}t,\ \ y_i = \int \psi_i(t)y(t)\,\mathrm{d}s,$$

x_j : expansion coefficients of x.